

Project Topic: Public Transportation

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Colab Link: [🔗](#) **Compiled Notebooks**

Literature Review/Introduction:

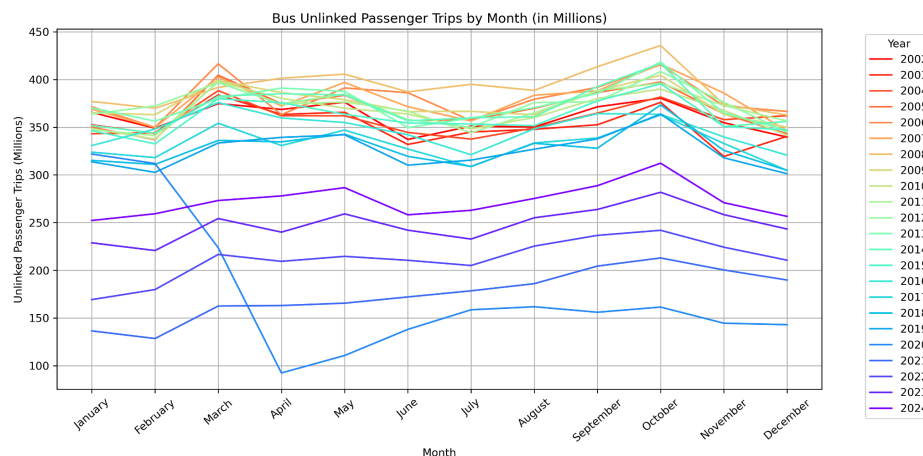
Public transportation is a defining feature of a community and plays a crucial role for providing mobility. Existing research suggests that quality public transportation is better for the public health of a community by providing access to healthcare and employment opportunities, reducing traffic congestion, and better for the environment [5, 7]. A lack of access to public transportation disproportionately affects those who rely on it, including elderly, low-income workers, and individuals with disabilities [9]. Additionally, accessibility to public transportation is necessary for equity, providing support especially for those with lower incomes [1]. Despite the value public transportation provides for a community, ridership and accessibility varies across the United States, with New York accounting for approximately 22% of bus ridership and 64% of rail ridership [12]. Ridership trends over time demonstrate that the coronavirus pandemic in 2020 brought public transit usage to an all-time low, with ridership slowly recovering since then.

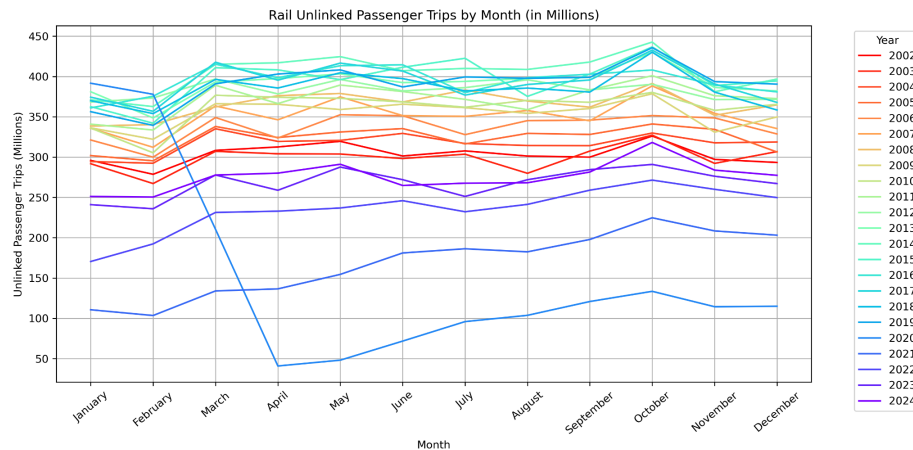
A major contributor to what affects ridership is who has accessibility to public transportation. Factors such as urban design, social vulnerability (poverty, disability, and minority status), local crime rates, and quality of transportation all play important roles. The relationship between urban design attributes—land use diversity, density, and street design—and public transportation accessibility has emerged as a critical issue in recent discussions regarding sustainable urban development. As such, effective urban design and public transportation planning are powerful and inclusive development tools, to mitigate growing urban problems such as public health, social inequity, road trauma and traffic congestion and environmental pollution [5, 6]. Areas with high social vulnerability, particularly those with higher poverty and minority populations, often face limited access to public transportation. National studies show that neighborhoods classified as transit deserts – areas where transit supply does not meet local demand – tend to have higher average levels of social vulnerability compared city-wide averages. On average, the social vulnerability in transit deserts is about 21 percent higher than in other neighborhoods [2]. Similar patterns have been observed in multiple U.S. regions, where communities with greater economic and social challenges face additional barriers to transit access, especially during crises [3,4]. Yet these trends do not always hold in each U.S. city. One study has found increased transit access among low income workers in Minneapolis and Minnesota at large compared to high income workers – a reversal of the national trend [14]. Another comes to the opposite conclusion when evaluating cities like New York and Houston, finding that high income white workers in these locales have more transit accessibility compared to low income non-white workers [15]. Thus,

this topic also warrants comparative analysis. Additionally, crime rates, especially violent and property crimes, significantly influence transit usage. Research by Jiao et al. [10] and Asgharpour et al. [11] shows that neighborhoods with higher crime rates often see lower ridership, particularly in low-income areas. Safety concerns lead residents to avoid public transit, and even with improved infrastructure, crime can suppress usage. These studies highlight the need to address both crime and safety alongside transit access to ensure equity in transportation. Investigating how these factors correlate with accessibility to public transportation can provide better insight on who is affected by lack of transportation.

Overview of Ridership Trends Over Time

We begin by examining transit ridership from 2002 to 2024 on a national level. Public transit ridership over time has fluctuated, especially during and after the COVID-19 pandemic. By visualizing unlinked passenger trips (the number of times passengers board public transportation vehicles) using data provided by the National Transit Database, we observe a decrease in ridership beginning in 2012, with a dramatic drop in ridership in April 2020 and a slow recovery to pre-pandemic levels in succeeding years. Previous research suggests competing modes of transportation, such as ride-hailing services, population density, and higher fares contribute to transit ridership change, although more recent research contradicts this, indicating socioeconomic factors, gasoline prices, telecommuting, and polycentric development collectively contribute more to post-pandemic ridership than vehicle revenue mile (VRM), fare, transit coverage, and service areas. [8, 12]. Additionally, public transit ridership seems to spike September - October each year, possibly due to changes in behaviors associated with changes in schedules at the end of summer.





Data Product: Marianne Finn

Although the trend suggests a decline in public transit ridership, especially by bus, compared to early 2000s, a lack of access to public transit may be exacerbating this. Nearly 45% of Americans have no access to public transportation options at all, denying people the option of choosing this mobility option [13]. In the following sections, we'll examine the role of public transit accessibility through its relationship with urban design, social vulnerability, local crime rates, and how access and quality varies across New York City, Minneapolis, San Francisco and Atlanta.

Relationship Between Urban Design and Public Transportation Accessibility

Many studies have shown that there exists a bidirectional interplay between individual urban design attributes—land use diversity, density, and street design—and transit. Neighborhoods with greater residential and business density, well-connected street networks, and access to a large variety of services see higher transit ridership and reduced car dependency [5, 6]. This section aims to assess whether these findings, when combined into a singular, composite score measure, apply and provide insights into the public transportation accessibility in NYC census block groups.

1. Land Use Diversity:

Mixed land uses generate overlapping trip purposes (e.g., living, working, shopping), reducing the need for long-distance travel and increasing the feasibility of walking or transit use.

Consequently, this creation of relatively self-contained societies reduces the need for internal urban mobility and renders the region easily serviced by public transportation. Studies highlight that neighborhoods with greater housing density, well-connected street networks, and access to a large variety of services see higher transit ridership due to proximity and reduced car dependency [1, 2].

- D2B_E8MIX : 8-tier employment entropy (denominator set to observed employment types in the CBG)
- D2R_JOBPOP: Regional Diversity. Standard calculation based on population and total employment: Deviation of CBG ratio of jobs/pop from the regional average ratio of jobs/pop

2. Density:

Population, housing and employment density are foundational factors in urban development. High-density residential and business areas concentrate potential riders, enabling frequent service and shorter headways, increasing the use of public transport [1]. Studies have found that communities with residential densities exceeding 20 dwellings per hectare correlates with increased walking, cycling, and transit use, as seen in cities like Melbourne [2]. High-density areas and intensive urban developments broaden the range of services accessible without resorting to automatic travel, reducing journey distances (average distances between homes, services, jobs, etc.) [1].

- D1A: Gross residential density (HU/acre) on unprotected land
- D1B: Gross population density (people/acre) on unprotected land
- D1C: Gross employment density (jobs/acre) on unprotected land

3. Street Design:

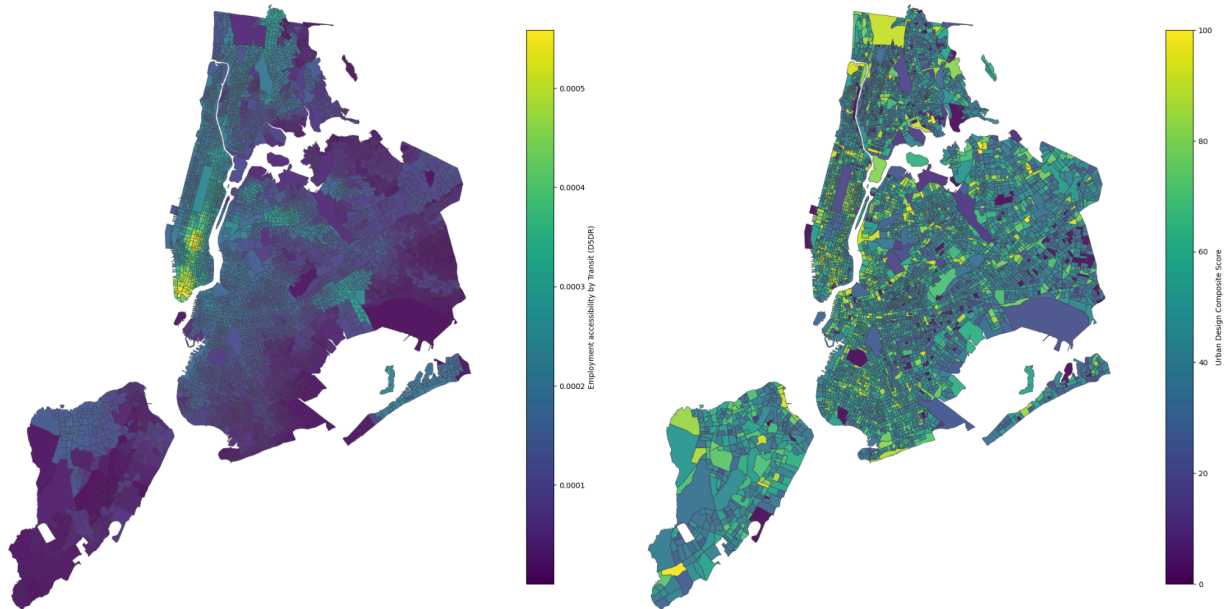
Pedestrian-oriented street networks—characterized by high connectivity, grid-like layouts, and sidewalk infrastructure—enhance the usage and accessibility of public transit. Small, dense street grids create a more accessible environment for pedestrians, reducing distances between land uses and vehicular speeds while fostering safer and more direct routes to transit nodes [1]. Conversely, curvilinear suburban layouts prioritize car travel, fragmenting pedestrian networks [1, 2]. The uses of intersecting streets make walking trips easier, reducing congestion at the entrances to residential areas.

- D5DRI: Regional Centrality Index of Transit
- NatWalkInd: Walkability Index
- D3A: Road Network Density
- D3B: Street intersection density (weighted, auto-oriented intersections eliminated)
- D4A: Distance from the population-weighted centroid to nearest transit stop (meters)

Note: For details regarding the methodologies involved in the computation of the Urban Design Composite Score, please refer to the submitted notebook file (brief written description are included).

Proportional Accessibility of Regional Destinations by Transit
(Employment accessibility by Transit, expressed as a ratio of total MSA accessibility)
in NYC CBGs

Urban Design Composite Score in NYC CBGs



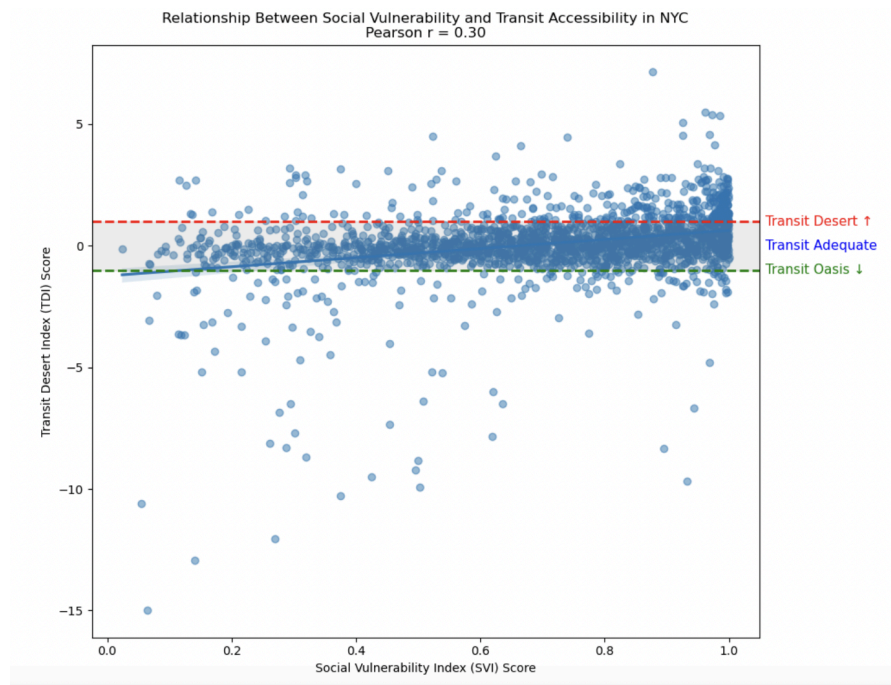
Data Product: Connie Chen

The first choropleth (D5DR: Transit Accessibility) shows the geographic distribution of employment accessibility by transit across census block groups in New York City. Darker colors indicate areas with lower transit accessibility, while lighter colors represent higher accessibility. The second choropleth (Composite Urban Design Score) displays the computed urban design quality composite score for the same census block groups, where higher scores (warmer colors) correlate with environments with more urban design attributes that promote employment accessibility by transit.

Relationship Between Social Vulnerability and Public Transit Accessibility

The relationship between social vulnerability and public transportation access is a critical concern for equitable urban development. Individuals considered socially vulnerable, such as those with low income, minority status, limited mobility, among others, often face greater barriers to essential services due to inadequate transit access. These disparities are quantified through tools like the Social Vulnerability Index (SVI) and Transit Desert Index (TDI), which aim to reveal inequities in urban transportation systems. National studies have demonstrated that areas with high vulnerability often coincide with poor transit service; over 24 million people live in transit deserts where the average SVI in these areas is over 20 percent higher than citywide averages [2]. Disparities are particularly pronounced in the New York-Newark-Jersey City metropolitan area where the SVI in transit deserts was nearly 50 percent higher than the regional average, with poverty rates similarly elevated [2]. Evidence from other U.S. regions further illustrates how inequities in transit access often mirror socio-economic divides. For instance, a

study in northeastern Illinois found that transit deserts disproportionately affect Hispanic, Black, and Asian communities, particularly in areas with overlapping vulnerabilities [3]. Additionally, service reductions during crises (e.g., COVID-19 pandemic) have most impacted groups with higher poverty rates and larger Black populations [4]. While these figures highlight the scale of inequity across the broader region, less is known about how these patterns manifest within New York City itself. This section addresses that by examining recent data to see how these patterns persist in a local context.



Data Product: Caroline Chan

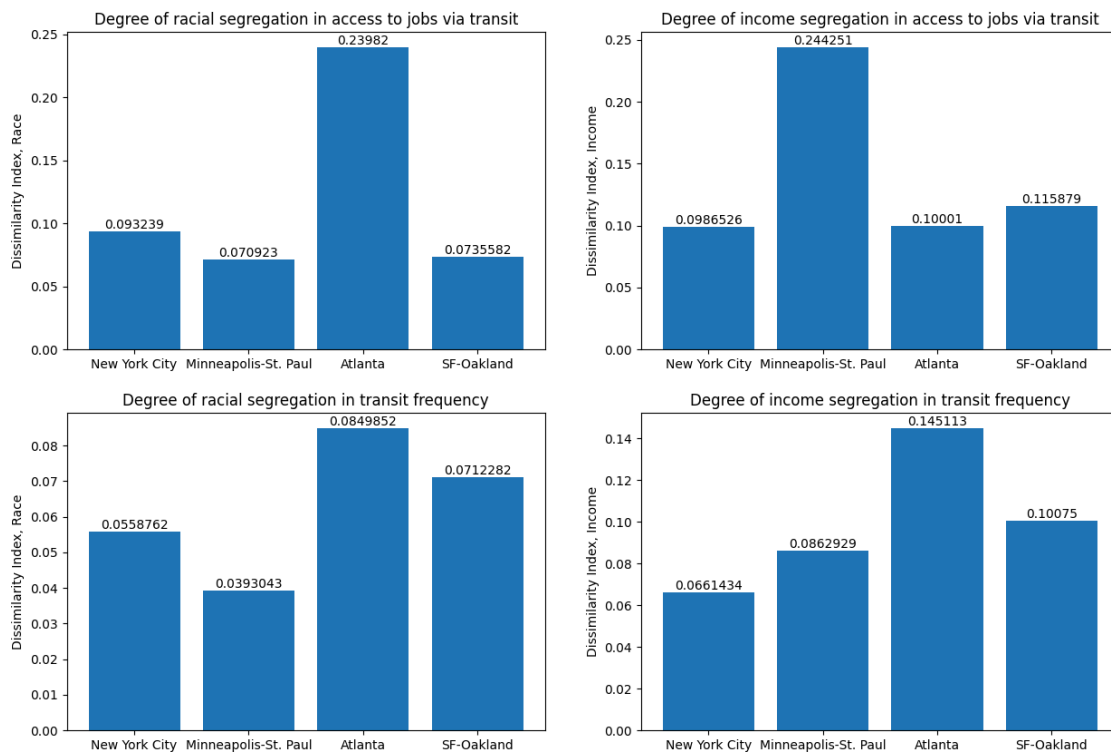
The scatterplot (Relationship Between Social Vulnerability and Transit Accessibility in NYC) shows the relationship between the Social Vulnerability Index (SVI) and the Transit Desert Index (TDI) across New York City census tracts. Each point represents a tract, with SVI on the x-axis and TDI on the y-axis. A weak to moderate positive correlation (Pearson $r = 0.30$) suggests that more socially vulnerable areas tend to have poorer transit access. However, the wide spread of points indicates that this relationship is not consistent across all neighborhoods. Some high-SVI tracts have adequate service, while some low-SVI areas still face transit challenges. This variation contrasts with earlier studies suggesting a stronger correlation, especially in the New York City metropolitan area [2]. This highlights the complexity of transit equity in New York City and may point to the need for more granular analysis (further explored in supplement).

Variance of Transit Accessibility Between US Cities

As a consequence of unique policy decisions and historical developments, US cities have each come to vary in their built environments – especially in regard to the accessibility and quality of

public transit. This section builds on the previous sections on transit accessibility and social vulnerability by approaching these topics through a comparative lens. Specifically, it will evaluate the degree to which transit access is segregated by race (Black vs. white) and income (above vs. below median) in four metropolitan areas: New York City, Minneapolis-St. Paul, Atlanta, and San Francisco-Oakland. These areas are meant to provide a broad survey of urban public transit options in the US. Finally, it will identify which cities, if any, are providing more equitable access to transit.

To quantify the degree to which transit access is segregated by some demographic characteristic in a given area, we have settled for the Dissimilarity Index computed over a three-way partition of the area's block groups into *low*, *medium*, and *high* transit accessibility categories. Dissimilarity index values are shown for each of the four cities below for both race and income in the figure below. In the first row of plots, we use the number of jobs accessible within a 45-minute transit commute from the given block group (D5BR) as the proxy for transit accessibility. In the second row, we change to using the aggregate frequency of transit service per capita for a given block (D4E).

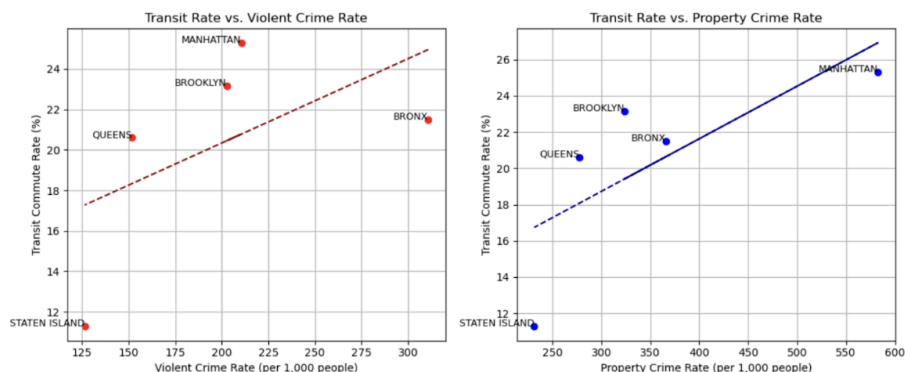


Data Product: Timothy Lin

In our findings, racial segregation in terms of transit frequency seems much less pronounced than racial segregation in terms of job access via transit. In contrast, the overall degree of segregation by income is roughly comparable for both transit accessibility proxies. Of note is the fact that public transit in Atlanta appears substantially more racially segregated than in the other cities. This finding aligns well with previous research on spatial segregation in Atlanta, inequalities in transit connectivity in Atlanta, and the distribution of the MARTA rail system [16]. Also of note is the prominence of Minneapolis-St. Paul in terms of income segregation in access to jobs via transit. This is because the areas of low job accessibility via transit are in the wealthy outer suburbs while the areas of high job accessibility via transit are in the substantially poorer interior (explored further in supplement). Overall, we find that New York City and San Francisco-Oakland-Berkeley have less transit segregation in terms of race and income.

Correlation of Local Crime Rates with the Use of Public Transportation

The relationship between local crime rates and public transportation usage has long been a topic of interest in urban studies. Crime, particularly violent and property crimes, may influence public transit ridership by affecting how safe individuals feel when using public transportation. This section explores the impact of crime on transit usage across New York City's five boroughs, comparing how violent crime and property crime rates correlate with the percentage of the population using public transportation. Preliminary analysis suggests that while higher crime rates are often associated with lower transit usage, the correlation is not necessarily causal. Factors such as population density, availability of alternative transportation options, and neighborhood-specific dynamics likely play a significant role. Using data from NYC's crime statistics and transit usage rates, we have plotted the relationship between violent and property crimes and public transportation ridership. This data highlights nuanced differences across boroughs, suggesting that the impact of crime on transit behavior varies depending on the type of crime and the socio-economic characteristics of the neighborhood.



Data Product: Amrith Samuel

Conclusion

Improvements in accessibility to public transportation may improve public transportation ridership, improving equity and health of the population. As the analysis of ridership trends indicates, factors such as the COVID-19 pandemic, economic changes, and telecommuting have had significant impacts on transit usage. The decline in ridership since 2020 highlights the challenges that public transportation systems face, but it also underscores the importance of improving access, particularly in areas that experience high levels of social vulnerability. Urban design, including land use diversity, density, and street connectivity, plays a key role in determining transit accessibility, with well-planned neighborhoods showing higher transit ridership and reduced car dependency.

Additionally, addressing issues of social vulnerability and crime is essential to promoting equitable transit access. Areas with high social vulnerability, often characterized by low income and minority populations, tend to face greater barriers to transit access. These challenges are exacerbated in transit deserts, where public transportation supply does not meet demand. Furthermore, local crime rates, particularly violent and property crimes, significantly impact public transportation usage. Safety concerns in high-crime areas often lead to reduced ridership, as residents may avoid public transit, especially in the evenings. Therefore, a comprehensive approach to improving public transportation must consider both infrastructure improvements and efforts to enhance safety, ensuring that vulnerable communities are not left behind and that transit services are accessible to all.

Citations:

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